

# **Newborough Warren**

## Hydrological Scenario Viewer

### **Reference Manual**

*Version 1.3 — April 2026*

Companion to: Hollingham (2026)  
*Journal of Hydrology: Regional Studies (submitted)*

scenario\_viewer.html — browser application  
[https://newbroman.github.io/Newborough\\_Hydrology/scenario\\_viewer.html](https://newbroman.github.io/Newborough_Hydrology/scenario_viewer.html)

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# 1. Overview and scope

The Newborough Warren Hydrological Scenario Viewer ([scenario\\_viewer.html](#)) is a self-contained, browser-based tool for exploring how the water table across the Newborough Warren coastal sand dune aquifer (Isle of Anglesey, North Wales) is predicted to respond to climate change and to forest-management interventions. The viewer runs entirely in the browser; no server, no internet connection after the initial page load, and no installation are required.

The viewer is an equilibrium tool. Given a scenario — a specified combination of rainfall, potential evapotranspiration (PET), canopy interception, and atmospheric-draw coefficients — it computes the steady-state change in water-table elevation at each of the 71 reference dipwells using the state-space model (SSM) coefficients fitted in Hollingham (2026) from 21 years of monthly monitoring. These per-well changes are then interpolated onto a map at 50 m resolution and displayed under a LiDAR-derived hillshade basemap.

Three questions are well-suited to the viewer:

- How does a future climate trajectory (UKCP18 2050s or 2080s, central estimate for Wales under RCP8.5) partition across the six hydrogeological clusters of the site?
- How does complete or partial removal of the Corsican pine plantation at the western edge of the site propagate through the C4 Forest cluster?
- Where do the ecological depth-to-water-table thresholds of Curreli et al. (2013) get crossed under a given scenario — i.e. which slacks are predicted to remain wet-slack viable, which to transition to dry-slack, and which to lose slack habitat altogether?

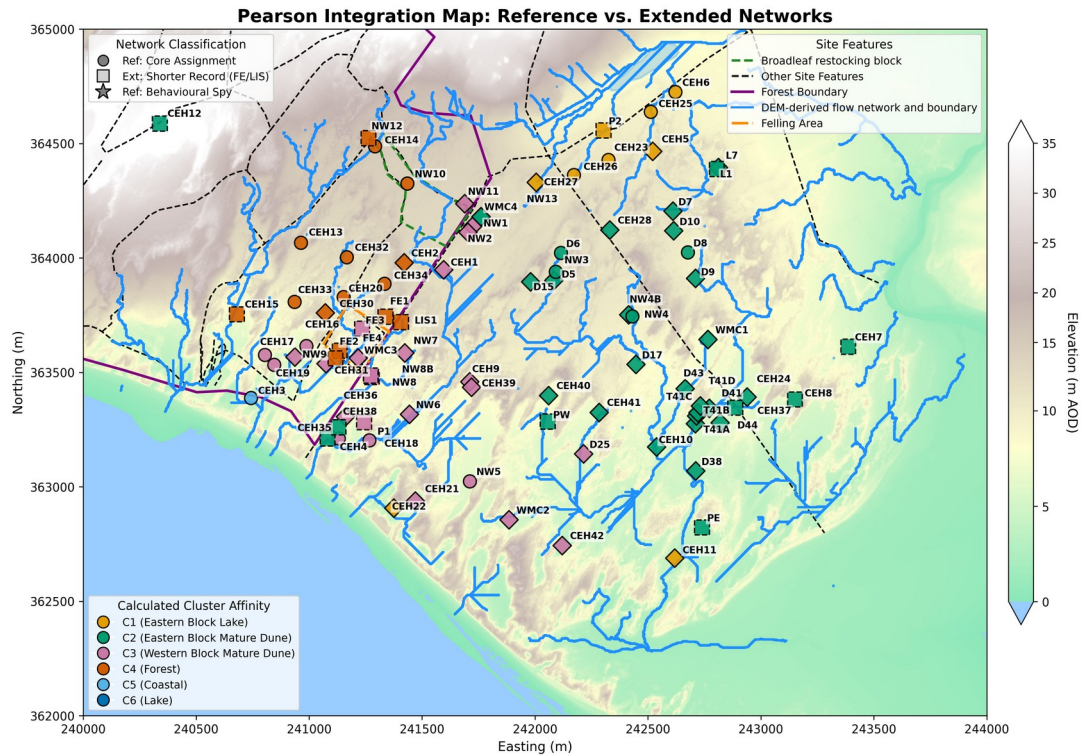
Three questions are outside the viewer's scope:

- Transient dynamics within a year. The viewer reports seasonal equilibria (Winter and Summer means), not month-to-month trajectories. For transient forecasting of specific future months, a dynamical SSM run is required.
- Cluster-to-cluster hydraulic propagation. The equilibrium framework is a per-well calculation; it does not simulate lateral groundwater flow between clusters. Scenarios that change forest-management parameters at C4 therefore do not propagate into C1 or C2 through the viewer, even though in reality some propagation may occur.
- Engineering or regulatory decisions. The viewer is a research communication tool; outputs should not be used as the basis for statutory decisions without independent verification.

**i** The viewer is built from scripts 01–19 of the Newborough Warren analytical pipeline, which is archived on GitHub ([https://github.com/newbroman/Newborough\\_Hydrology](https://github.com/newbroman/Newborough_Hydrology)) and on Zenodo. The fitted  $\beta$  coefficients, cluster assignments, and specific yield estimates that underpin the viewer are those published in the companion paper; the viewer is therefore fully reproducible from the pipeline.

## 2. The cluster system

Ward's-linkage clustering of the standardised monthly dipwell records (Hollingham, 2026, Section 4.1) identified six behavioural clusters across the 71-well network, five of which are monitored at a density sufficient to support the SSM fit. The viewer uses the five well-populated clusters (C1–C5); C6 (n=1) is retained in the reference data but does not appear on the scenario maps.



**Figure 1.** Spatial distribution of dipwells coloured by hydrogeological cluster membership. The frame matches the viewer extent (E 240200–243700, N 362400–364800, OSGB36).

## 2.1 Cluster characteristics

| Cluster                      | Position on site                | Key hydrological signature                                                                                                                                                                                                                                                    | n  |
|------------------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| C1 Eastern Block Lake-buffer | East of the site, lake-adjacent | Lake-buffered by Llyn Rhos-ddu; shallowest mean depth of the open dune clusters; high $\beta_3$ drainage rate; the most ecologically important cluster for winter flooding                                                                                                    | 8  |
| C2 Eastern Block Mature Dune | Central east and north          | High $\beta_1$ recharge sensitivity; classic mature-dune behaviour; flashy summer draw-down                                                                                                                                                                                   | 28 |
| C3 Western Block Mature Dune | Central west                    | Open dune; moderate $\beta_1$ and $\beta_2$ ; baseline for forest-management contrast                                                                                                                                                                                         | 22 |
| C4 Forest                    | Western plantation block        | Corsican pine canopy depresses effective recharge by ~24 % (Freeman, 2008); low $\beta_2$ reflects canopy interception shield; proximal to the rock ridge — CEH14 at the sand-ridge interface carries the largest lateral subsidy in the network ( $\alpha = +0.222$ m/month) | 9  |
| C5 Coastal                   | Southern coastal margin         | Sea-boundary dominated; largest drainage sink; low amplitude overall                                                                                                                                                                                                          | 1  |

The clusters are not merely descriptive; they are the unit of analysis throughout the paper and the viewer. Every figure and every scenario output in the viewer is organised by cluster, and the climate-sensitivity ranking  $C1 > C2 > C3 > C4 > C5$  that emerges in Section 4.9 of the paper reflects the inherited  $\beta$ -coefficient structure fitted at each well and averaged within the cluster.

## 2.2 Western and Eastern blocks

A practical shorthand used throughout the viewer and this manual is the division of the site into the Eastern Block (C1 + C2) and the Western Block (C3 + C4). The two blocks are distinguished by a subsurface till-and-sand contrast: the Eastern Block is a till-and-estuary-influenced dune aquifer with shallower sand and lower storage capacity, while the Western Block is a deeper, more uniform aeolian sand body. This geological contrast — rather than any surface ridge or barrier — is the fundamental division identified by the clustering analysis. The persistent vertical offset between C1/C2 and C3 (C3 running 0.3–0.4 m deeper on average despite broadly similar topographic position) reflects this storage-capacity contrast directly.

A separate geomorphological feature — the bedrock rock ridge — forms the north western boundary of the dune aquifer itself. The ridge is a linear SW–NE feature along the western edge of the site, rising from sea level in the south to over 50 m AOD in the north-western corner; ground elevations greater than 15 m AOD on the DEM correspond to the ridge proper rather than to any dune feature. The C4 Forest cluster sits in the northern part of the site, closest to the ridge, which is why CEH14 — at the edge where the Warren sands meet the rock ridge — is classified as C4 and carries the largest persistent lateral subsidy in the network ( $\alpha = +0.222$  m/month from the Model B intercept audit; Hollingham, 2026, Section 4.4). This subsidy feeds the C4 cluster, not the Eastern Block.



### 3. The state-space model in brief

The viewer is built on the state-space model (SSM) of Hollingham (2026). A full derivation, fit diagnostics, and cluster-level parameter tables appear in the companion paper; the description here is only what a practitioner needs to interpret viewer outputs with confidence.

#### 3.1 The core equation

At each dipwell, the change in water-table elevation ( $\Delta h$ ) from one month to the next is modelled as:

$$\Delta h(t) = \beta_1 \cdot P_{\text{eff}}(t) - \beta_2 \cdot PET(t) - \beta_3 \cdot h(t-1)$$

where  $P_{\text{eff}}$  is the effective rainfall reaching the water table after canopy interception,  $PET$  is Thornthwaite–Mather potential evapotranspiration, and  $h(t-1)$  is the well's depth-below-ground the previous month. The three  $\beta$  coefficients are fitted independently at each well and carry distinct physical meanings:

| Coefficient | Interpretation                                                                                         | Units     |
|-------------|--------------------------------------------------------------------------------------------------------|-----------|
| $\beta_1$   | Recharge sensitivity — how much the water table rises per unit of effective rainfall                   | mm / mm   |
| $\beta_2$   | Atmospheric-draw sensitivity — how much the water table falls per unit of PET                          | mm / mm   |
| $\beta_3$   | Internal-drainage brake — the intrinsic rate at which a raised water table declines toward equilibrium | per month |

The equilibrium (steady-state) head at a well is obtained by setting  $\Delta h = 0$  and solving for  $h$ . Under a scenario that perturbs  $P$  and  $PET$ , the change in equilibrium head,  $\Delta h_{\text{scenario}}$ , is computed directly from the  $\beta$  coefficients at each well. This is the quantity the viewer displays on the  $\Delta h$  vs baseline map.

#### 3.2 Specific yield

Specific yield ( $S_y$ ) is the fraction of the sediment volume that releases or absorbs water as the water table rises or falls. The viewer reports  $S_y$  alongside every scenario for display purposes, but — and this matters for interpretation —  $S_y$  does not appear in the  $\Delta h$  equation itself. The fitted  $\beta$  coefficients already embed the aquifer storage response through the SSM fit to observed head data; applying  $S_y$  as a divisor on  $\Delta h$  would be double-counting.

The viewer's  $S_y$  slider is a display scaling only. Moving it changes the reported Mean  $S_y$  and C4  $S_y$  values in the summary panel but does not change the  $\Delta h$  map.

**i** Why is  $S_y$  shown if it has no effect on  $\Delta h$ ? Because the viewer is often used for conversation with stakeholders, and  $S_y$  is a number practitioners expect to see quoted alongside any water-balance discussion. Hiding it would create more confusion than showing it.

## 4. The six scenarios

The viewer ships with six pre-configured scenarios. These are selected from the left-hand panel under the SCENARIOS heading and are mutually exclusive — only one scenario is active at a time. Any scenario can be further modified by the seasonal climate sliders (Section 5), the Forest interception slider, and the  $\beta_2$  scaling slider.

### 4.1 Baseline (2005–2026)

The reference state. All climate multipliers are 1.0 $\times$ , canopy interception is set to the Freeman (2008) Corsican pine value of 24 %, and  $\beta_2$  scaling is 1.0 $\times$ . The  $\Delta h$  map is uniformly zero under the baseline scenario —  $\Delta h$  is, by definition, the departure from baseline. To see absolute water-table elevations under baseline conditions, switch to the Absolute head display mode (Section 6).

### 4.2 UKCP18 2050s (RCP8.5, central estimate)

Applies the Met Office UKCP18 central-estimate (50th-percentile) projections for Wales under the RCP8.5 emissions pathway, at the 2050s time-slice (2040–2069). Four seasonal perturbations are applied:

- Winter (Nov–Mar) precipitation: +10 %
- Summer (May–Sep) precipitation: –15 %
- Winter PET: +5 %
- Summer PET: +20 %

These values match the scenarios applied in Section 4.9 of the main paper and are derived from the UKCP18 Regional 12 km projections (Met Office, 2018) as filtered through the CHESS-SCAPE bias-corrected projections of Robinson et al. (2023). The site-wide annual-mean response is  $\Delta h = -0.040$  m/month (equivalent to –0.48 m/year under sustained equilibrium), driven predominantly by the summer signal.

### 4.3 UKCP18 2080s (RCP8.5, central estimate)

Applies the same methodology as the 2050s scenario but at the 2080s time-slice (2070–2099), giving substantially stronger perturbations:

- Winter precipitation: +20 %
- Summer precipitation: –30 %
- Winter PET: +10 %
- Summer PET: +35 %

Site-wide annual-mean response:  $\Delta h = -0.072$  m/month (–0.86 m/year). The summer signal dominates; winter responses across all clusters are modestly positive (+0.015 m/month site-wide mean), reflecting the stronger fractional increase in winter precipitation than winter PET under UKCP18 for Wales.

**i** The UKCP18 perturbations used here are central-estimate values. The UKCP18 probabilistic range at the 10th and 90th percentiles spans roughly half to twice these central values for precipitation and a broader range still for PET. Users wishing to explore this envelope should use the seasonal sliders (Section 5), which allow independent perturbation of the four seasonal terms across the range 0.5 $\times$ –1.5 $\times$  of baseline.



#### 4.4 Clearfell (interception $\rightarrow 0$ , $\beta_2 \uparrow$ )

Simulates complete removal of the Corsican pine plantation across the nine C4 Forest cluster wells. Two parameter changes are applied simultaneously, only at C4 wells:

- Canopy interception drops from 24 % to 0 %, so that full rainfall reaches the aquifer.
- $\beta_2$  is scaled by 1.20 $\times$  at C4 wells, reflecting the BACI-observed post-felling ratio measured at the clearfell impact wells (Hollingham, 2026, Section 4.6.6).

Predicted  $\Delta h_{C4} = -0.024$  m/month annually, with a strongly asymmetric seasonal profile: summer  $\Delta h_{C4} = -0.050$  m/month (dominant), winter  $\Delta h_{C4} = +0.001$  m/month. The summer deepening reflects the sacrificial-shielding mechanism in which the intact canopy dissipates summer evaporative demand that post-clearfell falls directly onto the water-table store.

#### 4.5 Broadleaf conversion

Simulates replacement of the Corsican pine plantation with native broadleaf species. The canopy-interception fraction is reduced from 24 % (pine) to 15 %, following the temperate deciduous annual-mean meta-analysis of Komatsu et al. (2011);  $\beta_2$  is held at baseline on the basis that the annual-mean atmospheric-draw response is expected to be broadly comparable to pine once the summer-leafed and winter-leafless phenological cycle averages out.

Predicted  $\Delta h_{C4} = +0.006$  m/month annually (+0.07 m/year), reflecting the modest net recharge gain from reduced annual-mean interception. The seasonal asymmetry that underlies the phenological argument of Section 5.4.4 of the paper — larger winter recharge pulse under leafless canopy, steepened hydraulic gradient, accelerated drainage into summer — is a dynamical trajectory that the equilibrium framework does not resolve. The broadleaf scenario is therefore best understood as an annual-mean lower bound on the benefit.

#### 4.6 Forest thinning (50 %)

Simulates 50 % canopy thinning of the Corsican pine. Parameter changes are applied at half the clearfell magnitude: canopy interception drops from 24 % to 12 %, and  $\beta_2$  is scaled by 1.10 $\times$ .

Predicted  $\Delta h_{C4} = -0.012$  m/month annually, approximately half the clearfell response as expected from the linear structure of the equilibrium equation.

## 5. The seasonal climate sliders

Below the scenario selector, the CLIMATE (SEASONAL) panel exposes four sliders that allow independent perturbation of the four seasonal forcing terms:

- Winter P — scales November–March precipitation
- Summer P — scales May–September precipitation
- Winter PET — scales November–March potential evapotranspiration
- Summer PET — scales May–September potential evapotranspiration

All four sliders span the range 0.5×–1.5× of baseline in 0.05× steps. This range was chosen to bracket the UKCP18 end-century probabilistic projections for Wales under RCP8.5 at the 10th and 90th percentiles, while staying within the linear steady-state domain of the fitted state-space model. Moving a slider to 0.5× halves the baseline forcing in that season; moving to 1.5× increases it by 50 %.

### 5.1 Interaction with scenarios

The sliders are multiplicative on top of the active scenario. Selecting UKCP18 2050s sets the sliders to their scenario values (1.10×, 0.85×, 1.05×, 1.20×); the user can then move any slider to adjust that seasonal term while holding the others at the scenario values. This allows sensitivity analysis — for example, asking what the UKCP18 2050s response would look like if summer PET were even more severe than the central estimate suggests.

### 5.2 Annual aggregation

The summary panel reports the Mean  $\Delta h$  and C4  $\Delta h$  by default at the annual level. The annual value is computed as the 0.5-weighted mean of the winter and summer equilibrium  $\Delta h$  values:

$$\Delta h_{\text{annual}} = 0.5 \cdot (\Delta h_{\text{winter}} + \Delta h_{\text{summer}})$$

This convention — equal winter/summer weighting — preserves the seasonal asymmetry that is central to the viewer's findings. An annual-mean forcing (averaging the perturbations before running the SSM) would average out the contrasting winter and summer responses and obscure the dominant summer signal. The Annual/Winter/Summer tabs at the top of the summary panel allow the user to switch between aggregation schemes.

## 6. The three display modes

The top-right of the map panel has three mutually-exclusive display mode buttons:

- $\Delta h$  vs baseline — diverging red/blue scale showing the change in equilibrium water-table elevation from baseline
- Absolute head — sequential blue scale showing the water-table elevation in metres above ordnance datum (m AOD) under the current scenario
- Depth below surface — custom-anchored scale showing the per-cell depth from ground surface to water table, with Curreli thresholds marked

### 6.1 $\Delta h$ vs baseline

The default display. Uses a diverging colour scale centred at zero, with deeper red indicating stronger water-table deepening relative to baseline and deeper blue indicating stronger shallowing. This is the right view for communicating the relative impact of a management intervention or climate trajectory.

The colour scale range auto-adjusts to the current scenario so that the full range of  $\Delta h$  values is visible. Absolute  $\Delta h$  values greater than about  $\pm 0.05$  m/month are ecologically significant for dune slack vegetation; values greater than  $\pm 0.1$  m/month indicate a substantial change in habitat suitability.

### 6.2 Absolute head

Sequential blue scale showing water-table elevation in m AOD. This is the right view for showing the hydrological surface itself — the gradients, the divides, and the absolute position of the water table in space. The elevation high at the western edge of the site, corresponding to the bedrock rock ridge, is clearly visible in this mode, with the water table sloping away eastward and southward toward the coast.

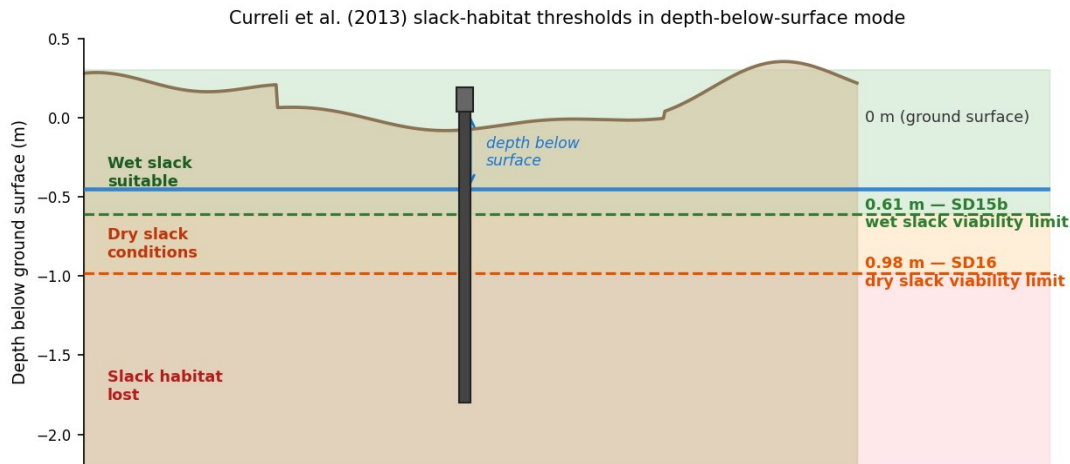
Absolute head values under baseline range from approximately 2 m AOD at the coastal margin to approximately 12 m AOD near the bedrock ridge on the western edge; values in the C1 Eastern lake-buffer cluster average around 9.5 m AOD. Under the UKCP18 2080s scenario the surface is lowered by roughly 0.2–0.8 m across most of the site.

### 6.3 Depth below surface

Computed per grid cell as:

$$d(E, N) = h_{DEM}(E, N) - h_{WT}(E, N)$$

where  $h_{DEM}$  is the bilinearly-interpolated LiDAR ground-surface elevation at the cell centre and  $h_{WT}$  is the IDW-interpolated scenario water-table elevation at the same cell. This is the right view for ecological interpretation: Curreli et al. (2013) define slack vegetation thresholds in depth-below-surface terms, and rendering the water table against the topography permits direct identification of slacks where scenarios cross these thresholds.



**Figure 2.** Depth-below-surface mode uses a custom colour scale anchored on the Curreli et al. (2013) slack-vegetation thresholds. The green band corresponds to depths shallower than 0.61 m (SD15b wet slack viability limit); the orange band is between 0.61 m and 0.98 m (SD16 dry slack viability limit); the red band is deeper than 0.98 m (slack habitat lost).

**i** Depth-below-surface mode applies the dune-ridge mask by default (see Section 8). On a dune ridge the interpolated water table sits several metres below ground, but this is a consequence of the ridge height rather than any hydrological feature, and rendering ridge cells would produce a misleading strong-red band along every dune crest. The mask is user-toggable from the MAP panel on the left.

## 7. Ecological thresholds (Curreli et al., 2013)

The viewer's depth-below-surface colour scale is anchored on the dune-slack vegetation thresholds of Curreli et al. (2013), derived from a multi-site analysis of dune slack plant communities across the British coastal dune system.

| Threshold          | Depth below surface | Ecological meaning                                                                                                                                           |
|--------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Winter flooding    | 0.10 m              | Above this depth the slack is seasonally inundated — prime wet slack habitat for specialist pioneer species                                                  |
| SD15b              | 0.61 m              | Wet slack viability limit. Summer minima deeper than 0.61 m begin to exclude the wet-slack community (SD15b association)                                     |
| SD16               | 0.98 m              | Dry slack viability limit. Summer minima deeper than 0.98 m exclude even the dry-slack community (SD16 association), transitioning toward dry-dune grassland |
| Slack habitat lost | 1.50 m              | Below this depth, the slack community is functionally absent; the viewer's depth scale saturates at deep red here                                            |

The viewer's depth colour scale places explicit tick marks on the legend at the SD15b and SD16 values so that scenario results can be read directly against the thresholds. The  $\Delta h$  and Absolute head modes do not display the Curreli thresholds — those thresholds are depth-below-surface-defined and do not map meaningfully onto elevation-based displays.

### 7.1 Why 40 cm appears in the paper but not the viewer

Readers of the companion paper will recognise a 37 cm threshold referenced in Section 4.1 in the context of annual-mean summer minima across clusters. This 37 cm value is the empirical difference in mean annual summer minimum depth between the wettest and driest slack communities observed at the site — an internal site statistic derived from the dipwell record. The Curreli thresholds (0.61 m, 0.98 m) are external, cross-site thresholds from the published literature. The two thresholds are complementary but not interchangeable: the 37 cm value tells us how much depth variation separates slack types at Newborough; the Curreli thresholds tell us what absolute depth exceeds the general ecological tolerance of each community.

The viewer uses Curreli rather than the 37 cm value because: (a) the viewer is intended to support scenario interpretation for readers outside the Newborough team, for whom published cross-site thresholds are more transferable; (b) the 37 cm difference is an annual-mean statistic, whereas the viewer operates at seasonal resolution; (c) the Curreli thresholds come with an established colour convention (wet slack green, dry slack orange) that is more intuitive for quick interpretation.

## 8. The dune-ridge mask

The dune-ridge mask is applied exclusively in depth-below-surface mode. Its rationale is straightforward: the water table is a continuous surface in space, and its elevation relative to ordnance datum is well-defined at every point within the site regardless of whether the overlying ground is a dune ridge or a slack floor. Elevation-based maps ( $\Delta h$  and Absolute head modes) therefore require no topographic masking.

Depth-below-surface, however, is meaningful only where the ground surface is approximately at the well-plane elevation — that is, on the slack floors where the monitoring dipwells sit. At a dune ridge the interpolated water table lies several metres below ground, but this is a consequence of the ridge elevation rather than any hydrological feature. Displaying such cells would produce a misleading strong-red band along every dune crest on the depth map, obscuring the ecologically relevant slack-floor response that is the whole point of the view.

### 8.1 How the mask is computed

The viewer embeds a 160×110 DEM elevation grid covering the viewer extent (E 240200–243700, N 362400–364800). For each display cell, the DEM elevation at that point is compared with the IDW-interpolated well-plane elevation (computed by the same IDW kernel applied to well DEM ground elevations rather than to water-table heights). A cell is masked in depth mode when the DEM elevation exceeds the well-plane IDW elevation by more than 1.0 m — i.e. the cell sits on a ridge that projects more than one metre above the surrounding slack-floor plane.

The 1.0 m threshold matches the ridge-mask convention applied to the static figures of the paper (`map_utils.add_idw_surface` default). The mask is user-toggleable from the MAP panel on the left of the viewer — tick or untick "Mask dune ridges (depth view)". Disabling the mask reveals the unmasked depth field, which is occasionally useful for diagnostic inspection but should not be used for ecological interpretation.

## 9. Map conventions — hillshade, tooltips, legend

### 9.1 Hillshade basemap

The map background is a LiDAR-derived greyscale hillshade rendered at approximately 35 % opacity underneath the scenario surface. It is generated from the Natural Resources Wales 2 m LiDAR DEM using standard Lambertian shading (illumination azimuth 315°, altitude 35°, vertical exaggeration 3), sampled onto a 1100×750 grid and embedded as base64 PNG in the HTML (approximately 200–300 KB). The hillshade is the same one used in the static figures of the paper; its purpose is to provide topographic context for scenario interpretation.

### 9.2 KML overlays

Three polygon overlays are drawn on top of the hillshade: the site boundary (purple outline), the Corsican pine plantation (green fill), and the Llyn Rhosddu freshwater lake (blue fill). These overlays can be toggled via the KML overlays checkbox in the MAP panel. Well symbols are drawn on top of all overlays and are toggleable independently via the Well labels checkbox.

### 9.3 Tooltips

Hovering over any well symbol reveals a tooltip showing: well name, cluster membership, baseline depth, scenario-perturbed depth,  $\Delta h$ , absolute head, and the well's fitted  $\beta$  coefficients. The tooltip provides a direct link between the per-well framework output and the ecological depth-below-surface interpretation, and is the quickest way to read specific numbers off a scenario without having to export the CSV.

### 9.4 Legend and colour bar

The legend appears below the map and updates with the active display mode. In  $\Delta h$  mode the legend is a diverging red–white–blue bar with min/max values shown; in Absolute head mode it is a sequential blue gradient; in Depth below surface mode it is the Curreli-anchored scale (green / orange / red) with explicit tick marks at 0.61 m and 0.98 m.

## 10. Interpreting results: what is signal, what is artefact

A few interpretive guidelines help separate the meaningful content of a viewer output from the artefacts of the equilibrium framework or the interpolation scheme.

### 10.1 Signal

- Directional differences between clusters. The climate-sensitivity ranking  $C1 > C2 > C3 > C4 > C5$  is robust and traces directly to the fitted  $\beta_1$  and  $\beta_2$  coefficients at each cluster. If a UKCP18 scenario shows C1 deepening more than C2, that ordering is physically grounded.
- Seasonal asymmetries. The finding that summer  $\Delta h$  is strongly negative under UKCP18 while winter  $\Delta h$  is modestly positive is driven by the fractional structure of the perturbations and the  $\beta_2$  response; it should be interpreted as a real prediction of the framework.
- Forest-management effect confined to C4. Under clearfell, thinning, and broadleaf scenarios, the response is spatially confined to the C4 cluster and its immediate IDW-interpolated surroundings. This is correct: the framework modifies  $\beta$  coefficients only at C4 wells, and the BACI monitoring record (Hollingham, 2026, Section 4.6) empirically confirms that canopy effects do not propagate significantly into the open dune clusters.

### 10.2 Artefacts

- Extrapolation beyond the well network. Near the coastal margin and in the south-east corner, the power-1 IDW kernel extrapolates beyond the convex hull of the wells. The triangulation-linear method used for the published figures correctly leaves these areas blank; the viewer extrapolates under the IDW kernel for rendering continuity. Values in these extrapolated zones should be treated as indicative only.
- Sub-interpolation-scale features. Features smaller than the inter-well spacing (~100–300 m) cannot be resolved. Localised till lenses, perched water tables, or small drainage channels that are not sampled by the monitoring network will not appear on the map.
- Annual-mean broadleaf response. The broadleaf conversion scenario predicts a small annual-mean response because  $\beta_2$  is held at baseline; the phenological mechanism (larger winter recharge pulse under leafless canopy) is not resolved by the equilibrium framework. The broadleaf  $\Delta h$  values should therefore be treated as a lower bound on the benefit, not as a calibrated prediction.



## 11. Known limitations

| Limitation                                      | Practical implication                                                                                                                                                                                                                |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Equilibrium, not dynamical                      | The viewer reports steady-state seasonal equilibria, not within-year trajectories. Carry-over effects from winter storage into summer minima are embedded only in the fitted $\beta_3$ but not explicitly resolved.                  |
| $\beta$ coefficients fixed at historical values | Future climate may alter the $\beta$ structure itself (e.g. higher $\beta_2$ under consistently warmer summers). The viewer holds $\beta$ fixed and does not simulate structural shifts.                                             |
| No lateral coupling                             | The per-well equilibrium framework does not simulate cluster-to-cluster groundwater flow. Management actions at C4 therefore cannot propagate into C1 or C2 through the viewer, even where lateral propagation may occur in reality. |
| Central-estimate UKCP18                         | The UKCP18 2050s and 2080s presets use central-estimate (50th-percentile) projections only. The probabilistic envelope is substantially wider; users should explore it via the seasonal sliders.                                     |
| Monthly-resolution SSM                          | The underlying SSM is fitted to monthly-mean dipwell data. Short-duration flooding peaks or flashy drawdown events cannot be represented.                                                                                            |
| 71-well reference network                       | Spatial resolution is limited by the well density (~100–300 m inter-well spacing). Features below this scale cannot appear on the map.                                                                                               |

Addressing the first three limitations would require a calibrated continuous-flow model (e.g. MODFLOW), which was evaluated during development and judged to overclaim relative to the available hydraulic-conductivity and aquifer-thickness data at the site. A detailed discussion of the evaluation and rejection of this approach is given in Supplementary Note S3 of the paper.

## 12. Glossary and reference tables

### 12.1 Glossary

| Term                        | Meaning                                                                                                                      |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------|
| $\Delta h$ (delta-h)        | Change in water-table elevation between a scenario and the baseline (metres)                                                 |
| $\beta_1, \beta_2, \beta_3$ | Fitted SSM coefficients governing recharge, atmospheric draw, and internal drainage                                          |
| BACI                        | Before–After–Control–Impact experimental design used in the clearfell evaluation                                             |
| IDW                         | Inverse-distance weighting — the spatial interpolation method used by the viewer                                             |
| LCSC                        | Linearised catchment storage coefficient — the mm of rainfall required to raise the water table by 10 cm at a well           |
| m AOD                       | Metres above Ordnance Datum (approximately mean sea level)                                                                   |
| OSGB36                      | Ordnance Survey of Great Britain 1936 — the coordinate reference system used throughout                                      |
| PET                         | Potential evapotranspiration — the atmospheric demand for water computed by the Thornthwaite–Mather method                   |
| $P_{\text{eff}}$            | Effective rainfall reaching the water table after canopy interception; $P_{\text{eff}} = P \times (1 - \text{interception})$ |
| RCP8.5                      | Representative Concentration Pathway 8.5 — a high-emissions climate trajectory                                               |
| $S_y$                       | Specific yield — fraction of sediment volume that releases or absorbs water per unit water-table change                      |
| SSM                         | State-space model — the autoregressive equation fitted at each dipwell                                                       |
| UKCP18                      | The UK Climate Projections 2018 set, maintained by the Met Office                                                            |
| WTF                         | Water table fluctuation method — used to estimate specific yield from dipwell records                                        |

### 12.2 Cluster quick reference

| Cluster | n wells | $\beta_1$ (mm/mm) | $\beta_2$ (mm/mm) | $\beta_3$ (month <sup>-1</sup> ) | R <sup>2</sup> | Cluster label             |
|---------|---------|-------------------|-------------------|----------------------------------|----------------|---------------------------|
| C1      | 8       | 1.56              | 4.98              | 0.366                            | 0.66           | Eastern Block Lake-buffer |
| C2      | 28      | 1.67              | 4.38              | 0.200                            | 0.60           | Eastern Block Mature Dune |
| C3      | 22      | 1.20              | 3.59              | 0.118                            | 0.50           | Western Block Mature Dune |
| C4      | 9       | 0.96              | 3.41              | 0.078                            | 0.46           | Forest                    |
| C5      | 1       | 0.98              | 1.47              | 0.017                            | 0.27           | Coastal                   |

Values shown are cluster-mean approximate  $\beta$  values taken from 03\_03\_cluster\_mechanistic\_coefficients.csv. Individual wells within a cluster vary around these means; exact per-well values are visible via the well tooltips in the viewer.

## 12.3 RAF Valley monthly PET climatology

Long-run monthly medians of Thornthwaite–Mather PET at RAF Valley (~16 km from Newborough), used by the SSM as the atmospheric-demand input. The baseline climate is the 2005–2026 monthly mean.

| Month    | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|----------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| PET (mm) | 5   | 9   | 32  | 46  | 73  | 88  | 92  | 79  | 46  | 22  | 8   | 4   |

## 13. References

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